cl_px4_mr::CbLoiter
+ CbLoiter()
+ CbLoiter()
+ setParams()
+ params()
+ setCenter()
# buildPath()
# behaviorName()
cl_px4_mr::CbPx4PathFollowerBase
+ CbPx4PathFollowerBase()
+ ~CbPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
# buildPath()
# behaviorName()
# onPathStarted()
# onPathCompleted()
# totalLength()
# progressFraction()
# carrotVertexIndex()
- commandFor()
- applyCentering()
- tangentYawAt()
cl_px4_mr::CbPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
optional< std::chrono ::steady_clock::time _point >
-watchdogStart_
#goalChecker_
#localPosition_
#trajectorySetpoint_
#offboardKeepAlive_
#vehicleStatus_
#vehicleCommand_
#tunnelCentering_
vector< cl_px4_mr:: NedPoint >
vector< float >
atomic< size_t >
-path_
-cumLen_
-carrotVertex_
cl_px4_mr::FlightPattern LoiterParams
cl_px4_mr::PathFollowerParams
cl_px4_mr::NedPoint
+direction
-lastProgressDecile_
+yawMode
+altitudeAgl
+centerX
+centerY
+radius
+arrivalXyTol
+arrivalZTol
+autoTimeoutFactor
+fixedYaw
+groundSpeed
+leash
+minSegmentLength
+x
+y
+yaw
+z
-active_
-completed_
-entryHeading_
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ SmaccAsyncClientBehavior()
+ ~SmaccAsyncClientBehavior()
+ SmaccAsyncClientBehavior()
+ operator=()
+ SmaccAsyncClientBehavior()
+ operator=()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
-onEntryThread_
-onExitThread_
-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_
-onFailure_
-onFinished_
-onSuccess_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ ISmaccUpdatable()
+ operator=()
+ ISmaccUpdatable()
+ operator=()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-periodDuration_
-lastUpdate_
cl_px4_mr::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
-heading_
-refAlt_
-x_
-y_
-z_
-projection_
+onLocalPositionReset_
+onPositionReceived_
-localPosition_
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
-goalX_
-goalY_
-goalZ_
-xyTolerance_
-zTolerance_
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
-lastSetpoint_
-trajectorySetpoint_
cl_px4_mr::CpOffboardKeep Alive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
cl_px4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
+onArmed_
+onDisarmed_
+onLanded_
+onModeChanged_
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
cl_px4_mr::TunnelCentering Params
+ minRangeM
+ wallBandHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing
+ceilingClearanceM
+columnHalfWidthM
+floorClearanceM
+floorLookBackM
+lateralSearchM
+lookAheadMaxM
+lookAheadMinM
+maxLateralM
+maxVerticalM
+minFloorClearanceM
...
-haveAttitude_
-firstMessage_
-initialized_
+queueSize
+postInitialMessageEvent
+postMessageEvent
smacc2::client_core _components::CpTopicSubscriber < MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
smacc2::client_core _components::CpTopicSubscriber < sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
< sensor_msgs::msg ::PointCloud2 >
cl_px4_mr::CpTunnelCentering
+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
- onCloud()
- onAttitude()
atomic< int64_t >
array< float, 4 >
atomic< float >
time_point
-params_
-lastCloudNs_
-attitudeQ_
-ceiling_
-floor_
-lateral_
-left_
-right_
-vertical_
-subscriber_
-lastLog_
-timeoutMs_
-lastUpdateTime_
smacc2::ISmaccStateMachine
smacc2::introspection ::SmaccStateInfo
smacc2::ISmaccState
smacc2::ISmaccOrthogonal
smacc2::ISmaccClientBehavior
smacc2::ISmaccClient
smacc2::ISmaccComponent
smacc2::SignalDetector
cl_px4_mr::MapProjection
smacc2::SmaccSignal